



The Ultimate Tree & BST Master Super-Notes



Complete Revision Guide • Concepts + Visual Diagrams + Memory Connections + Interview Secrets

1. The Foundation: Linear vs Non-Linear World

📌 Story Memory Hook: Socho ek seedhi single-lane sadak (Array/Linked List) jahan aage ya peeche hi jaa sakte ho. Lekin **Tree** ek multi-branch magical ped hai jahan har mod par alag-alag raste khulte hain!

Property	Linear Data Structures (Array, LL)	Non-Linear Data Structures (Trees)
Path	Single sequential path (0 to n ya node.next)	Hierarchical (Multiple branching paths)
Structure	1D Sequential / Flat	Parent-Child levels (Org Chart, Folders)
Traverse Direction	Sidha aage ya peeche	Naturally Top-to-Bottom. Wapas aane ke liye <i>Recursion/Call Stack</i> lagta hai!



Core Tree Terminology (Bache Jaisa Simple Dictionary)



Root Sabse upar wala Main Boss node (Level 0). Iska koi parent nahi hota.



Leaf Node Tree ke aakhiri bache jinka koi child nahi hota (Degree = 0).



Internal Node Root aur Leaf ke beech wale nodes jo aage branch failate hain.



Siblings Ek hi mummy-papa (Parent node) ke bache jo same level par hote hain.



Levels & Height Root = Level 0, Uske bache = Level 1, Pote = Level 2. Tree ki gehrayi = Height.

2. Classification & Types of Trees (The Family Members)

Tree Type	Golden Rule / Pehchan	Real-world Memory Link
Binary Tree	Har node ke Maximum 2 bache (0, 1, ya 2).	"Hum do, humare do" rule.
Binary Search Tree (BST)	Left < Root < Right har sub-level par.	Smart library jisme sorted placement hoti hai.
Full / Strict Binary Tree	Har node ke paas Exactly 0 ya 2 bache honge. (1 bacha strictly forbidden ❌).	Strict rule: Ya toh bilkul nahi, ya poore 2.
Complete Binary Tree	Saare levels completely packed, aur aakhiri level Left-to-Right fill hona chahiye.	Cinema hall seating: pehli rows full, last row left se bharte jao.
Skewed Tree	Saare nodes ek hi line mein (Sirf Left ya Sirf Right).	Tree jo bigad kar wapas Linked List ban gaya!

Degenerate Tree	Har level par sirf 1 node (chahe seedha ho ya zig-zag).	Single chain formation.
Extended Binary Tree	Incomplete tree mein Dummy/Null nodes add karke full banana.	Khali seats par dummy putle bitha dena.

3. Memory Representation (Computer Tree Ko Kaise Save Karta Hai?)

Option A: Array-Based (Formula Magic)

Tree ko Level-by-Level flat array mein store karo index i par:

- **Left Child:** $2 * i + 1$
- **Right Child:** $2 * i + 2$
- **Parent:** $\text{floor}((i - 1) / 2)$

⚠ **Bada Nuksan:** Agar tree skewed ya khali ho toh bohot saare index khali chhut-te hain = Memory Barbaad!

Option B: Linked List-Based (Standard)

Har node 3 kamro (compartments) ka ek block hota hai:

[left_ptr | DATA | right_ptr]

- Dynamic memory allocation (Jab chahiye tab banao).
- Zero memory wastage. Khali jagah par None point karta hai.

4. The 3 Tree Travelers (Traversal Techniques)

🧠 **The Master Trick:** Traversal ka naam is baat par hai ki humare **CAPTAIN ROOT** ki baari kab aati hai!

Note: Left bacha hamesha Right bache se pehle visit hoga! (L → R rule fixed hai).

🏆 **Pre-Order (Root Pehle):** ROOT → Left → Right

Use Case: Tree ki clone/copy banana, Prefix expression evaluation.

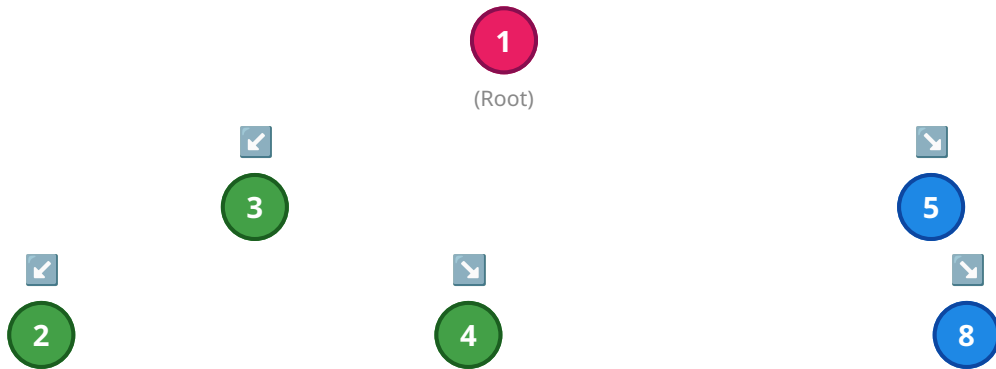
🥈 **In-Order (Root Beech Mein):** Left → ROOT → Right

Use Case: BST ka sorted (ascending) order output nikalna!

🥇 **Post-Order (Root Aakhiri):** Left → Right → ROOT

Use Case: Bottom-up deletion (Tree delete karna bina orphan chhode), Directory folder size.

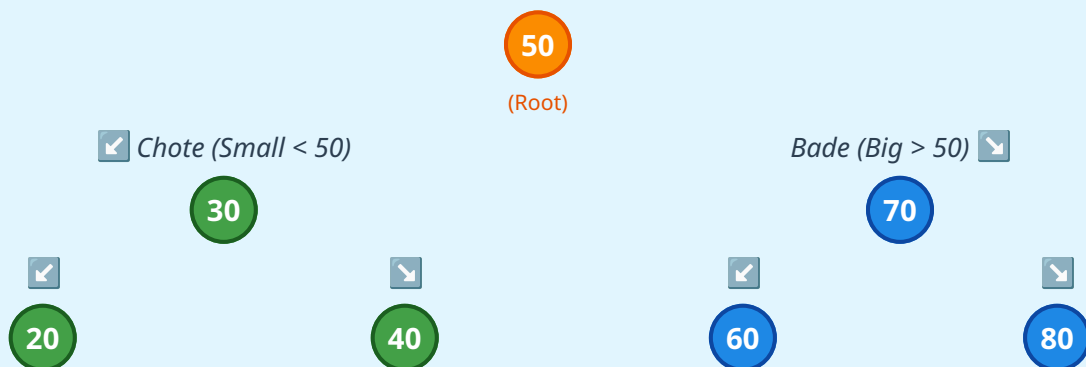
🎨 Visual Traversal Example (Demo Tree)



Preorder (R-L-R): 1, 3, 2, 4, 5, 8 | **Inorder (L-R-R):** 2, 3, 4, 1, 5, 8 | **Postorder (L-R-R):** 2, 4, 3, 8, 5, 1

5. Binary Search Tree (BST) Deep-Dive

🎯 **BST Kyu Banaya? (The Speed Benefit):** Normal binary tree me element dhoondhne ke liye har node visit karna padta hai = $O(n)$ (Slow, bilkul linked list jaisa). BST me hum har step par aadha tree skip karte hain = $O(\log n)$ (Super Fast!).



🔍 Searching Logic:

1. Current node se compare karo.
2. Chota hai → Go Left.
3. Bada hai → Go Right.
4. None mila → Element Not Found!

✚ Insertion Logic:

1. Search ki tarah rasta follow karo.
2. Jahan aage rasta khatam (None mile), wahan naya node bana ke attach kar do!
3. Time: $O(\log n)$ per node.

6. 🧠 Grand Memory Link & Interview Golden Secrets

🎯 The 3-in-1 Interview Master Trick

Interviewer aapko ghumane ke liye 3 alag-alag sawal pooch sakta hai:

1. "BST ke sare elements ko increasing (sorted) order mein print karo."
2. "Kaise check karoge ki given tree ek Valid BST hai ya nahi?"
3. "Ek BST ka In-order Traversal code likho."

💡 **TEENO KA ANSWER EK HI HAI: IN-ORDER TRAVERSAL!** 💡

Kyunki BST ki golden property hai: **In-order (Left → Root → Right) run karne par output HAMESHA strictly Sorted Ascending Order aata hai!**

Upar wale BST tree ka In-order: 20, 30, 40, 50, 60, 70, 80 (Sorted ✅).

⚡ 10-Second Quick Recap ⚡

Tree = Non-linear Hierarchy • Binary Tree = Max 2 kids • Full = 0 ya 2 kids • Complete = Left-to-right packed • Skewed = 1 line (slow) • Array Formula = $2i+1$, $2i+2$ • Pre = Root pehle, In = Root beech, Post = Root aakhiri • BST = Left < Root < Right • BST Inorder = Sorted Magic!